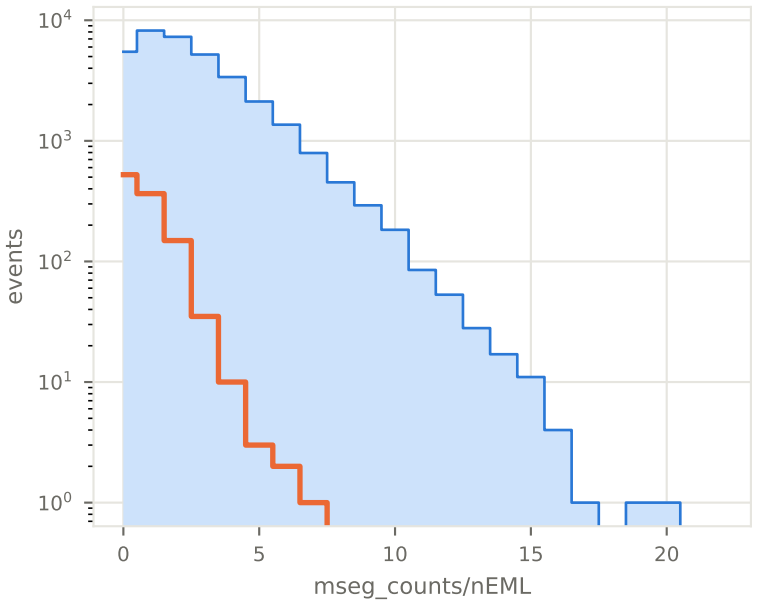
mseg_counts/nEML

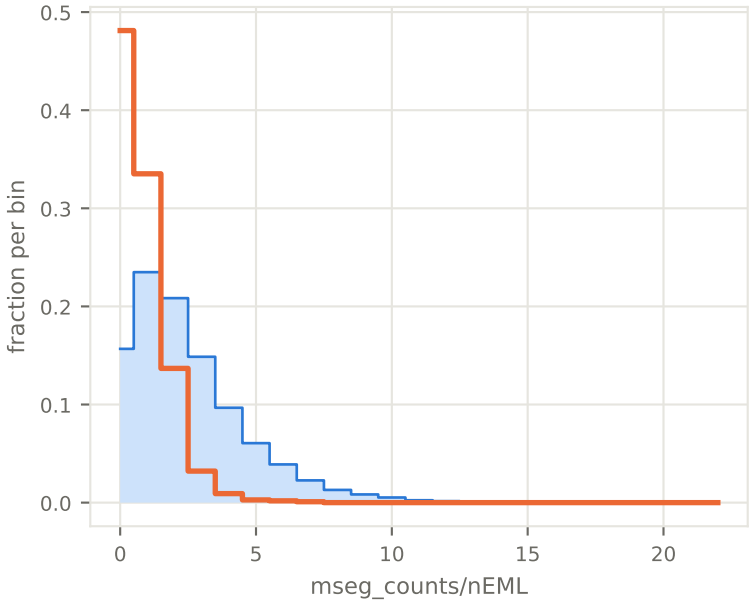
BARREL vs ENDCAP · shape · rank 53 / 98 by |D_barrel - D_endcap| · neither

shape D(barrel vs endcap): background 0.425 signal 0.437 separation D: endcap 0.033 barrel 0.090

1a · background (data) · raw

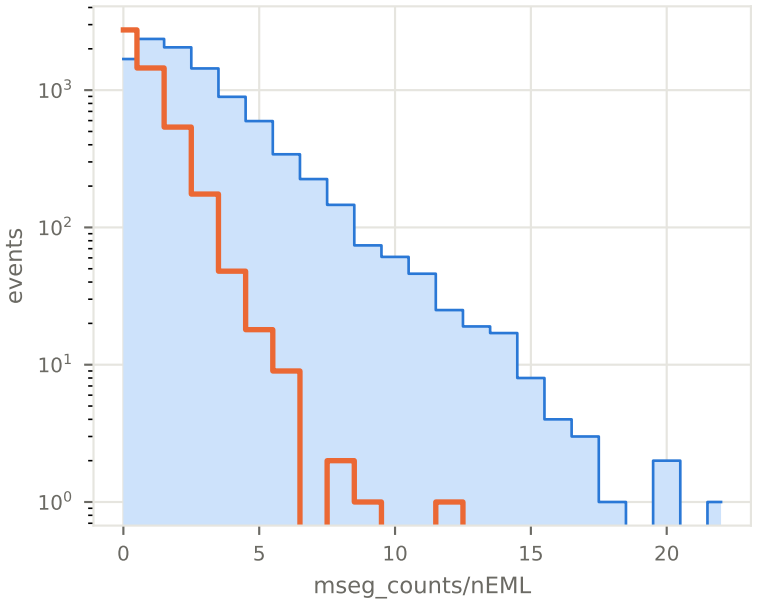


1b · background (data) · normalized

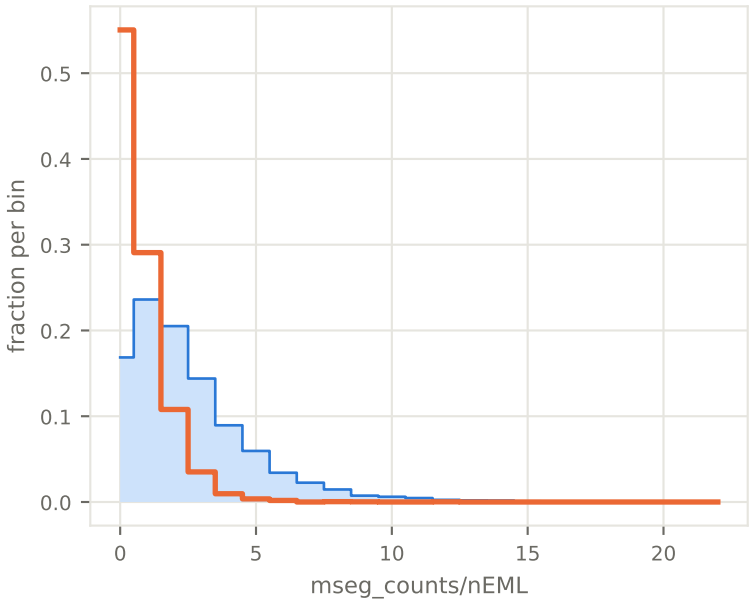


1 · background (data)
shape D = 0.425
endcap n 34,970
barrel n 1,089

2a · combined signal (MC, pooled) · raw



2b · combined signal (MC, pooled) · normalized



2 · combined signal (MC, pooled)
shape D = 0.437
endcap n 9,996
barrel n 4,987

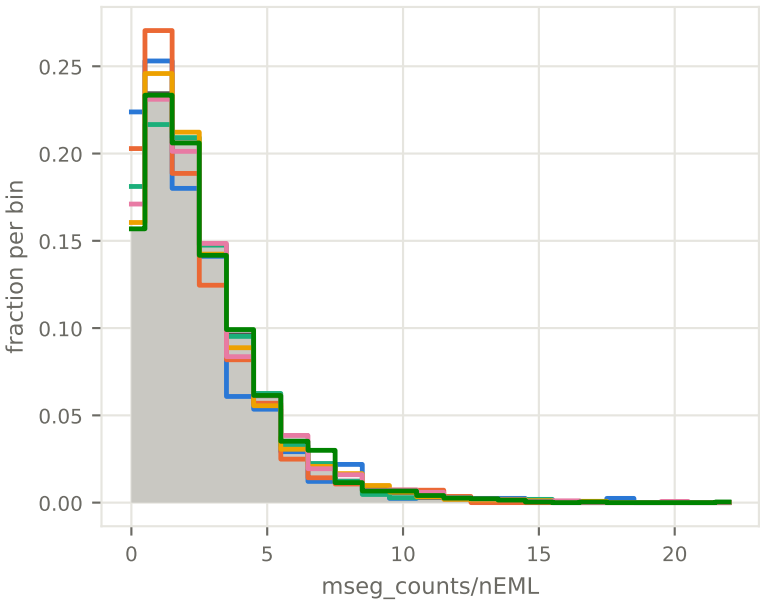
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

mseg_counts/nEML

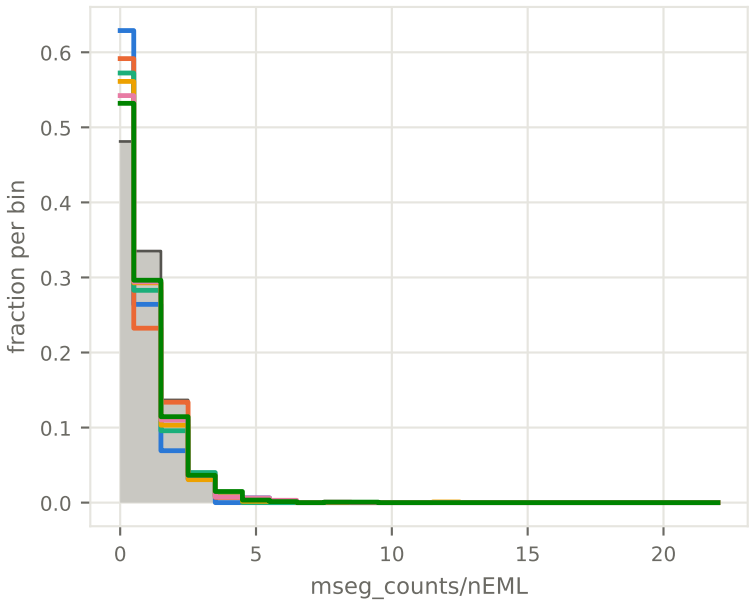
BARREL vs ENDCAP · separation side by side · rank 53 / 98 · neither

D_equal endcap 0.033   barrel 0.090   difference (barrel - endcap) +0.057   (normalized only: raw counts differ ~30x between regions)

1a · ENDCAP · each mass point vs background

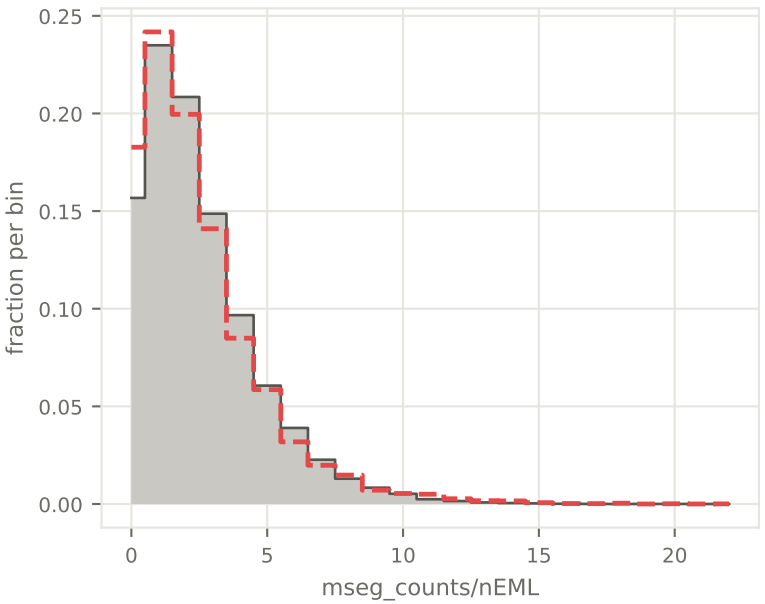


1b · BARREL · each mass point vs background

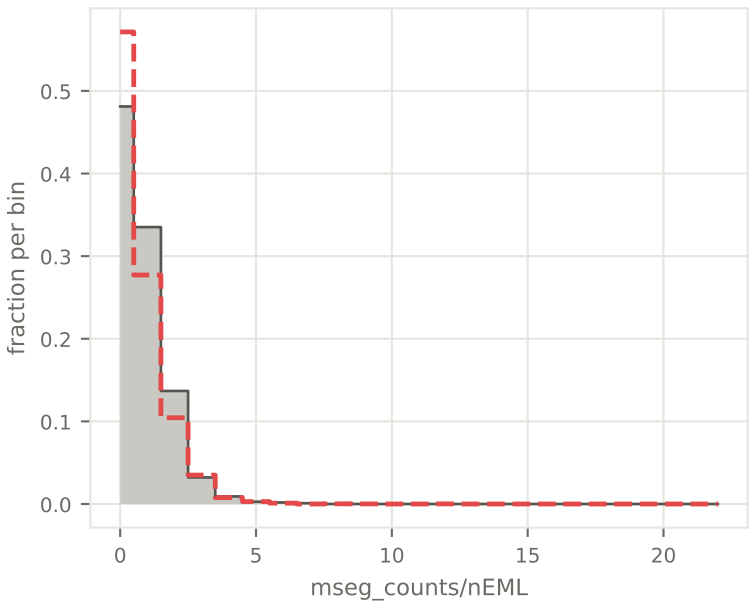


1 · each mass point
D: endcap | barrel
background n 34,970 | 1,089
mH125_mS15 D 0.085 | 0.148
mH125_mS55 D 0.082 | 0.110
mH200_mS50 D 0.024 | 0.091
mH400_mS100 D 0.018 | 0.080
mH600_mS150 D 0.014 | 0.061
mH1000_mS275 D 0.011 | 0.051

2a · ENDCAP · combined signal vs background



2b · BARREL · combined signal vs background



2 · combined signal
D: endcap | barrel
background n 34,970 | 1,089
combined, equal weight D 0.033 | 0.090

a = endcap b = barrel · normalized · edge bins include under/overflow · NaN and -999 removed